Filippo Delzanno

Problem Set #1 (CAAM 37711)

(1) a - We let Victor's utility function be the amount of money that he has. Supposing his entire bankroll is d, then the amount Victor can bet on Biden is a number $b \in [0, d]$.

- we can suppose that a market pricing Biden at $\mathbb{P}(\text{Biden wins}) = 0.6 \equiv q$ will sell contracts for $\$q$ paying $1 if Biden wins. With $1, one can buy $1/q$ contracts that pay out $(1\$ \times \frac{1}{q}) = \frac{1}{q}\$$. The total payout in this market is thus:

$$\text{profit} \equiv \text{payout} - \text{stake} = \frac{1}{q} - 1 = \frac{1-q}{q} \quad \text{for } q = 0.6$$

Thus, we can develop a utility function for Victor that describes the value of betting $\$b$ on Biden winning, given the probability with which he believes Biden will win. Letting $X \equiv$ Biden wins the election:

$$U(b, X) = \begin{cases} d + \frac{(1-q)}{q} b & ; \text{ if Biden wins, } \mathbb{P}(X) = 0.85 \\ d - b & ; \text{ if Biden loses, } \mathbb{P}(\neg X) = 0.15 \end{cases}$$

- the expected utility under this function for market odds $q = 0.6$ is:

$$\mathbb{E}[U(b,X)] = (0.85)\left[d + \frac{(1-0.6)}{0.6} b\right] + (0.15)[d - b] = d + \frac{5}{12} b$$

Since Victor is a rational agent, he will maximize their expected utility under:

$$b^* = \underset{b}{\operatorname{argmax}}\ \mathbb{E}_{X \sim p}[U(b,X)] = \underset{b \in [0,d]}{\operatorname{argmax}} \left(d + \frac{5}{12} b\right) = d,$$

which is maximized for $b^* = d$, since the expected utility is a strictly increasing function of b. Thus, we have shown that if Victor's utility function is the amount of money that he has, he will bet his entire bankroll on Biden winning.

This kind of linear utility of money does NOT necessarily make sense. The suggestion that one should spend everything they have on a bet with 15% chance of failing will likely be viewed as irrational, since losing everything is a stronger detriment than maximizing wins. Similarly, following this rule on every favorable bet will eventually lead to complete ruin, since the probability of winning n repeated bets at $\mathbb{P}(X) = 0.85$ goes like $(0.85)^n \to 0$ for large n. This is a problem for this principle, which suggests you bet everything each time, eventually leading to a complete loss. More broadly, strict adherence to this linear utility principle is not sensible, since it would avoid any decision that incurs a negative expected value. Hedging by buying insurance, for example, is a typical decision by rational agents that would incur negative expected value, contradicting the principle of linear utility of money.

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(1) b - Now, let's suppose that Victor's utility from y is given by utility function:

$$U(y) = \log y$$

- We can write the bet b as a fraction of Victor's wealth $f \in [0,1]$ for convenience, such that his total net worth after the bet is:

$$\text{Net worth} = \begin{cases} d + \frac{1-q}{q} b = d + \frac{1-q}{q} fd = d\left(1 + \frac{1-q}{q} f\right) & \text{; if Biden wins} \\ d - b = d - fd = d(1-f) & \text{; if Biden loses} \end{cases}$$

- Victor's expected utility under this scheme becomes:

$$\mathbb{E}[U(f,X)] = 0.85 \log\left[d\left(1 + \frac{1-q}{q} f\right)\right] + 0.15 \log[d(1-f)]$$

$$= 0.85 \log d + 0.85 \log\left[\left(1 + \frac{1-q}{q} f\right)\right] + 0.15 \log d + 0.15 \log(1-f)$$

$$= \log d + \left\{0.85 \log\left[\left(1 + \frac{2}{3} f\right)\right] + 0.15 \log(1-f)\right\},$$

where in the last step, we substitute $q = 0.6$ from the betting markets. A rational agent will choose to maximize their expected utility, so we can optimize $\mathbb{E}[U(f,X)]$ to get the fraction f of his total wealth that Victor should commit to the bet:

$$\frac{dE}{df} = \frac{0.85\left(\frac{2}{3}\right)}{1 + \frac{2}{3} f} - \frac{0.15}{1-f} = 0 \quad \rightarrow \quad \frac{17}{30}(1 - f^*) = \frac{3}{20}\left(1 + \frac{2}{3} f^*\right)$$

$$\rightarrow \boxed{f^* = \frac{5}{8}}$$

$$\left.\frac{d^2 E}{df^2}\right|_{f = f^*} = \left[-\frac{17}{45(1 + \frac{2}{3}f)^2} - \frac{0.15}{(1-f)^2}\right]\Bigg|_{f = \frac{5}{8}} \approx -1.25 < 0$$

$$\rightarrow \left.\frac{dE}{df}\right|_{f=0} = \frac{17}{30} - \frac{3}{20} = \frac{5}{12}$$

$$\rightarrow \left.\frac{dE}{df}\right|_{f=1} \rightarrow -\infty \text{ as } f \rightarrow 1$$

$\Rightarrow f^*$ is the global maximum on $f \in [0,1]$

Thus, to maximize his expected utility under $U(y) = \log y$, Victor should bet $b = \frac{5}{8} d$ on Biden winning.

c - I probably would not make the bet, on the basis of the uncertainty between the poll numbers and the market odds. Without at least one data point to discriminate whether the spread in probability is a result of a hidden variable or a genuine market mispricing, I would be hesitant to risk 60% of my net worth on a bet that has probability within 15 - 40% of failing. With this uncertainty, I would feel more secure betting 25 - 50% of the amount calculated in part (b) at the expense of smaller gains. If I found additional information that drops the spread in probability of losing and narrows on ~ 15% (or less), I would be more confident in applying the bet at full size.

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(2) Let's suppose that Y is binary with outcome $Y \in \{0,1\}$. The predictive function f assigns $x \in X$ a number $f(x) \in [0,1]$ that corresponds to the predicted conditional probability $\mathbb{P}[Y=1 \mid X=x]$. From lecture, we know that the cross-entropy loss will be:

$$\mathcal{L}(y,p) = -[y \log p + (1-y)\log(1-p)]$$

- we can denote the TRUE conditional probability by a function $\eta(x) = \mathbb{P}[Y=1 \mid X=x]$. To show that the cross-entropy loss is a proper scoring rule, we must simply show that the risk is minimized when $f^*(x) = \eta(x)$, i.e.:

$$f^* \equiv \underset{f}{\operatorname{argmin}}\ \mathbb{E}_{x,y}[\mathcal{L}(Y, f(x))] \quad \text{where} \quad f^*(x) = \eta(x) = \mathbb{P}[Y=1 \mid X=x]$$

Pf | Let's compute the expected loss for a fixed value of x:

$$\mathbb{E}_{x,y}[\mathcal{L}(Y,f(x))] = \mathbb{E}_x\left[\underbrace{\mathbb{E}[\mathcal{L}(Y,f(x)) \mid X]}\right] \quad [\text{Tower property}]$$

for fixed x, $p = f(x)$ and $Y(x) = \eta$ such that $Y=1$ with probability η and $Y=0$ with probability $1-\eta$

$$= \mathbb{E}_x\left[-(\eta \log p + (1-\eta)\log(1-p))\right]$$

→ the overall risk is the average of the expected loss at fixed x over the distribution of X

- call this term $g_x(p) \equiv -(\eta \log p + (1-\eta)\log(1-p))$. Since the value of $p = f(x)$ at a fixed point x only affects this function, we should choose the $f(x)$ that minimizes $g_x(p)$ at every x. No other choice of $f(x)$ can do better than this, so this will also be the optimal minimizer in the overlying $\mathbb{E}_x[g_x(p)]$. We can minimize this function for $0 < \eta < 1$:

$$\frac{dg_x(p)}{dp} = -\frac{\eta}{p} + \frac{1-\eta}{1-p} = 0 \quad [\text{optimization condition}]$$

$$\rightarrow \eta(1-p) = p(1-\eta) \rightarrow p^* = \eta \quad [\text{there is only 1 critical point}]$$

$$\left.\frac{d^2 g_x(p)}{dp^2}\right|_{p^*} = \left[\frac{\eta}{p^2} + \frac{1-\eta}{(1-p)^2}\right]\Bigg|_{p^*=\eta} = \frac{1}{\eta} + \frac{1}{1-\eta} > 0 \quad \forall\ \eta \in (0,1)$$

* at the endpoints $p = 0, 1$: $g_x(p)\big|_{p\to 0} \to \infty$ AND $g_x(p)\big|_{p\to 1} \to \infty$

Thus, the critical point $p^* = \eta$ is the unique minimum on $0 < \eta < 1$. Now, we should check $\eta = 0$ and $\eta = 1$, since these are possible values for the true conditional probability:

- if $\eta = 0$, $g_x(p) = -\log(1-p)$, which has its minimum at $p = 0 = \eta$] since $p \in [0,1]$
- if $\eta = 1$, $g_x(p) = -\log(p)$, which has its minimum at $p = 1 = \eta$

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(2) Thus, in every case, the value of p that minimizes $g_x(p)$ is $p = \eta(x)$. This implies that, for every x, the expected loss at that x is uniquely minimized by taking:

$$p^* = f^*(x) = \eta(x)$$

- Since we showed that minimizing the expected loss at every x will minimize the implied risk, we can conclude that the predictor function that minimizes the implied risk is

$$f^*(x) = \eta(x) \equiv \mathbb{P}[Y \mid X = x]^{\dagger}$$

† initially, we defined $\eta(x) \equiv \mathbb{P}[Y = 1 \mid X = x]$, but since Y is binary, the entire conditional distribution $\mathbb{P}[Y \mid X = x]$ is specified by knowing $\eta(x)$. So they are interchangeable.

Thus, the cross-entropy loss is a proper scoring rule in the case where Y is binary. ☒

(3) The convergence condition $\sum_t \rho_t^2 < \infty$ that we discussed in class will guarantee that stochastic gradient descent (SGD) will converge to the minimum of the objective function. In a finite-sample setting, we can typically interpret the empirical risk

$$\hat{R}_n(w) = \frac{1}{n} \sum_{i=1}^{n} \mathcal{L}[w; x_i, y_i]$$

as an objective function. For this objective, the minimizer $\hat{w}_n$ is only a noisy estimate of the optimal parameters of the population. This implies that there is some intrinsic noise between the true optimum and $\hat{w}_n$ that cannot be reduced by refining your optimization step. Thus, once your optimization error falls below this threshold, you cannot improve your test performance by increasing the precision of your SGD step.

By instead selecting a constant step size, we can perform a SGD that rapidly converges toward $\hat{w}_n$ and then fluctuates around it in a neighborhood whose size likely scales with the step size ρ and some method-intrinsic constant. If we choose ρ such that the neighborhood of fluctuation about $\hat{w}_n$ is smaller than the noise threshold discussed above, we reach the highest accuracy we are capable of much faster than if we were to decay the step size.

Further, we can imagine that a finite-sample setting could host conditions in which settling at the absolutely true optimal parameters would require knowledge about the training set that does not reflect the real trend that produced the data. We would have spent considerable resources to derive parameters that are unnecessary (or worse, obfuscating). Thus, violating the formal convergence criteria can be worthwhile in practice in order to reach an approximately acceptable solution very quickly.